

CRYSTAL CHENG

cjcheng@berkeley.edu | [linkedin.com/in/chengjcrystal](https://www.linkedin.com/in/chengjcrystal) | github.com/chengjcrystal | chengjcrystal.vercel.app

EDUCATION

University of California, Berkeley, Berkeley, CA

Expected Dec 2026

Bachelor of Arts in Computer Science & Applied Mathematics

- Relevant Coursework: Artificial Intelligence, Machine Learning, Statistical Modeling, Linear Algebra, Probability Theory, Discrete Math, Numerical Analysis, Data Structures, Computer Architecture
- Involvements: Society of Women Engineers (Treasurer), Web Development at Berkeley (Developer), Berkeley Project (Developer)

WORK EXPERIENCE

Hyve Solutions | Software Engineer Intern

June 2026 - Present

Software Engineer Team in Manufacturing & Integration

Newrium Foundation Inc | Software Engineer Intern

Sep 2025 - Dec 2025

- Owned development of customer-facing support and FAQ workflows, translating Figma designs into production-ready Next.js features across a 15-person cross-functional team
- Shipped 15+ reusable React/TypeScript components, reducing implementation overhead and establishing scalable frontend architecture patterns across the platform
- Architected MySQL schemas storing 100+ records and developed a Node.js email automation pipeline with SQL task queues, collaborating on API contract design to enable reliable end-to-end feature integration

Barobo | Software Engineer Intern

May 2025 - Aug 2025

- Blocked unauthorized video sharing across 100% of user traffic by building a single-use link system using AWS CloudFront signed URLs with expiration-based access control
- Added AWS DynamoDB to track user access metadata, scaling the backend to handle 10,000+ video share requests
- Reduced average video load time from ~800ms to ~760ms by adding CloudFront edge caching in front of S3 retrieval, a 5% drop that compounds across 10,000+ monthly requests
- Rewrote deployment as reusable Terraform and AWS CLI scripts, bringing provisioning time down from 2 hours to 15 minutes

Opal Mentorship App | Machine Learning Intern

May 2024 - May 2025

- Raised mentor-mentee match accuracy by 12% across cohorts by tuning 6 algorithmic scoring weights over two rounds of iteration
- Built a repeatable evaluation pipeline in AWS Lambda and DynamoDB that scored compatibility across 4 behavioral dimensions, giving the team a consistent basis for weighting decisions at small cohort scale
- Created a JavaScript scoring dashboard visualizing compatibility distributions across cohorts, driving 2 rounds of weight calibration

Swingbeats Research Lab | Frontend Web Developer

Jan 2024 - May 2024

- Increased user engagement with haptic feedback research products by redesigning the responsive website UI in Figma and implementing it with React, HTML, and CSS
- Refactored the homepage into reusable React components, cutting redundant render logic across 3 page sections

PROJECTS

Project Name | MongoDB, Express, Node.js, JavaScript

[link](#)

- Raised mentor-mentee match accuracy by 12% across cohorts by tuning 6 algorithmic scoring weights over two rounds of iteration
- Built a repeatable evaluation pipeline in AWS Lambda and DynamoDB that scored compatibility across 4 behavioral dimensions, giving the team a consistent basis for weighting decisions at small cohort scale
- Built a JavaScript scoring dashboard to visualize compatibility distributions across cohorts, informing two rounds of weight re-calibration

Project Name | MongoDB, Express, Node.js, JavaScript

[link](#)

- Raised mentor-mentee match accuracy by 12% across cohorts by tuning 6 algorithmic scoring weights over two rounds of iteration
- Built a repeatable evaluation pipeline in AWS Lambda and DynamoDB that scored compatibility across 4 behavioral dimensions, giving the team a consistent basis for weighting decisions at small cohort scale

SKILLS

- Languages: Python, C/C++, Javascript, SQL, Bash, HTML/CSS (Tailwind), MATLAB
- Machine Learning & Frameworks: PyTorch, Sklearn, NumPy, Pandas, React, Next.js, REST APIs
- Cloud, DevOps, & Tools: AWS (S3, CloudFront, DynamoDB, Lambda), Git/Github, Linux, Docker, Jenkins, Terraform, Postman